

Article

Comparative Analysis of Different AI Approaches to Stroke Patients' Gait Analysis

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Abstract

Despite advances in diagnostics, the objective and repeatable assessment of patients with neurological deficits (e.g., stroke) remains a major challenge. Modern methods based on artificial intelligence (AI) are of interest to researchers and clinicians in this area. This study presents a comparative analysis of different AI approaches used to analyze gait of stroke patients using a retrospective dataset of 120 individuals. The main objective is to evaluate the effectiveness, accuracy, and clinical relevance of machine learning (ML) and deep learning (DL) models in identifying gait abnormalities and predicting rehabilitation outcomes. Multiple AI techniques—including support vector machines (SVM), random forests (RF), k-nearest neighbors (k-NN), and convolutional neural networks (CNN)—were trained and tested on time-series gait data with spatiotemporal parameters. Performance metrics such as accuracy, precision, recall, and area under the curve (AUC) were used to compare model results. Initial results indicate that DL models, particularly CNNs, outperform traditional ML methods in capturing complex gait patterns and providing reliable classification. However, simpler models showed advantages in interpretability and computational efficiency. This study highlights the potential and shortcomings of AI-based gait analysis tools in supporting clinical decision-making and planning personalized stroke rehabilitation.

Keywords: artificial intelligence; machine learning; gait parameters; clinical gait assessment



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1. Introduction

Despite advances in diagnostics, the objective and repeatable assessment of patients with neurological deficits (e.g., stroke) remains a major challenge. Traditional qualitative gait analysis relies on observational gait testing and is subjective and largely dependent on the observer's clinical experience. Quantitative gait analysis, on the other hand, relies on parameter measurements with good accuracy and repeatability, which translates into more reliable diagnosis and comparative assessment during rehabilitation, even with less experienced diagnosticians/therapists [1]. Modern methods based on artificial intelligence (AI) are of interest to researchers and clinicians in this area [2].

The origins of clinical gait analysis based on artificial intelligence (AI) can be traced back to the long-standing use of biomechanics and motion capture to study human locomotion. Early gait analysis relied on visual observation and simple video recordings, which

lacked precision and objectivity [3]. With the development of force plates, marker-based motion capture, and electromyography, clinicians gained access to detailed biomechanical data, although their interpretation remained complex and time-consuming [4–7]. Advances in machine learning have created opportunities to automate data processing and extract clinically relevant patterns [8–10]. Artificial neural networks (ANNs) and support vector machines (SVMs) were first used to classify gait disorders and distinguish between normal and abnormal gait patterns [11]. The advent of deep learning enabled the analysis of raw sensor or video data without the need for extensive manual feature extraction [12]. Wearable sensors, such as inertial measurement units, enabled continuous and real-time gait monitoring, further enhancing the value of AI methods [13]. By combining multimodal data sources, AI models have improved diagnostic accuracy, rehabilitation monitoring, and personalized treatment planning [14]. Wearable technologies and AI (primarily machine learning (ML)) are gaining increasing attention in gait research, but their clinical application is not yet well-established and will not be widely available anytime soon [15]. These tools, combined with telerehabilitation, represent a paradigm shift in the analysis and assessment of post-stroke gait, providing tools for acquiring, storing, and analyzing multivariate, complex gait data with their nonlinear dynamic variability [16]. They also offer significant benefits such as automatic gait classification (e.g., hemiplegic gait), predicting future health status in this area, and analyzing trends and potential for reversal if unfavorable [1]. Recently, explainable AI techniques have been introduced to increase trust and transparency in clinical decision-making [17]. Currently, AI-based gait analysis is emerging as a powerful tool for early disease detection, assessing treatment outcomes, and providing access to healthcare beyond specialized laboratories [18,19].

Artificial intelligence (AI)-based clinical gait analysis in rehabilitation and physical therapy for neurological patients, including those recovering from stroke, requires accurate and reliable motion capture systems to quantify spatiotemporal and kinematic parameters [20]. Integration with wearable sensors, such as inertial measurement units or compression pads, enables continuous monitoring, both in the clinic and at home [21]. The system must handle noisy or incomplete data, as patients may have irregular or compensatory movement patterns [22]. Robust AI algorithms are essential for classifying gait abnormalities, tracking progress, and predicting recovery trajectories [23]. Interoperability with electronic medical records and rehabilitation devices ensures personalized treatment planning and outcome assessment [24]. User-friendly interfaces are essential so that clinicians can easily interpret results without requiring extensive technical training. Real-time feedback capabilities can support motor learning and corrective strategies during therapy sessions [25]. Data privacy and security are crucial given the sensitive nature of patient health information [26]. The system must be adaptable to a wide range of neurological conditions, severity levels, and assistive devices, ensuring inclusiveness [27]. To maintain accuracy, reliability, and clinical utility, validation through clinical trials and continuous model updates is essential [28,29].

AI-based clinical gait analysis in rehabilitation and physical therapy for neurological patients, including stroke patients, faces several research gaps that hinder its widespread clinical implementation. These include

- Limited validation in diverse populations due to any AI models have been developed and validated in controlled settings with small sample sizes, often characterized by a lack of diversity in age, gender, and ethnicity. This limits their generalizability to broader patient populations [30].
- Lack of standardization because a lack of consensus on standardized protocols for data collection, feature extraction, or model evaluation leads to inconsistencies across studies and hinders comparison of results or their integration into clinical practice [31].

- Data privacy concerns because the use of wearable sensors and video recordings poses significant privacy issues, especially for sensitive medical data. Ensuring data security and patient consent is a priority but often insufficiently addressed.
- Interpretability of AI models since many AI systems act as “black boxes,” providing predictions without clear explanations. This lack of interpretability can hinder clinician trust and acceptance, as understanding the rationale behind decisions is crucial to decision-making [32].
- Integration with clinical processes because AI tools often operate in isolation and are not seamlessly integrated with existing clinical processes or electronic health record systems, limiting their utility in real-world settings [33].
- Practical application: most AI models are tested in controlled environments, but their effectiveness in real-world settings, such as patients’ homes or community clinics, remains understudied [34].
- Challenges with longitudinal monitoring because continuous gait monitoring over extended periods is essential to assessing rehabilitation progress, but current AI systems may lack long-term tracking capabilities or be impractical for daily use [35].
- Multimodal data integration: Integrating data from various sources, such as video, wearable sensors, and clinical assessments, poses challenges in data fusion and analysis but is essential for comprehensive gait analysis [36].
- Personalization of interventions because AI models often provide generalized conclusions without tailoring recommendations to individual patient needs, which is crucial for effective rehabilitation [37].
- Regulatory and ethical issues due to implementing AI in clinical settings raises regulatory concerns about safety, effectiveness, and ethical issues that are still being addressed by policymakers and regulators.

Bridging these gaps is essential for the successful integration of AI-based gait analysis into rehabilitation and physical therapy for neurological patients, ensuring the effectiveness, reliability, and benefits of these tools for patients.

Clinical gait analysis based on artificial intelligence (AI) in rehabilitation and physical therapy for neurological patients, including stroke patients, faces several challenges that limit its full clinical potential. Accurately capturing gait patterns in patients with severe motor impairments is challenging due to compensatory movements and performance variability [38]. Integrating heterogeneous data from wearable sensors, motion capture systems, and video recordings requires advanced preprocessing and synchronization [39,40]. AI models often struggle with small datasets, as collecting large amounts of high-quality patient data is time-consuming and expensive [41]. Ensuring model generalization across different populations and stages of recovery remains a significant hurdle [42,43]. Many AI systems operate as “black boxes,” making it difficult for clinicians to interpret and trust their results. Real-time analysis and feedback are computationally intensive, especially when using multiple sensors or high-resolution video recordings [44]. Adherence and usability in home rehabilitation are challenging, as patients may find devices cumbersome or difficult to use. Regulatory approvals and ethical considerations, including data privacy and informed consent, complicate clinical implementation. Seamless integration of AI tools into existing clinical processes and electronic health record systems is often lacking. Continuously updating and validating AI models to reflect evolving patient populations and treatment protocols remains a continuing challenge.

The aim of this article is to provide a comparative diagnostic analysis of different artificial intelligence approaches to gait analysis in stroke patients.

The novelty and contribution of this paper lies in its systematic evaluation of multiple AI techniques applied to a highly complex medical problem. Unlike previous studies that

focused on a single algorithm, this work highlights the strengths and weaknesses of different models in capturing specific gait abnormalities caused by stroke. It introduces a direct comparison framework that emphasizes accuracy, interpretability, and clinical relevance, which are crucial in real-world rehabilitation settings. By comparatively analyzing the performance of different datasets and patient profiles, it identifies the most appropriate AI methods for personalized gait assessment. This study contributes to the development of precision rehabilitation by guiding researchers to AI tools that are both technically robust and clinically relevant.

The results presented in this article are only part of a larger study on artificial intelligence-based clinical assessment of gait analysis and the effects of neurorehabilitation in stroke patients, including

- Methods for rapid and easy-to-automate gait analysis;
- New gait parameters, including fuzzy and fractal logic;
- Methods for objectifying the assessment of gait rehabilitation effectiveness.

2. Materials and Methods

This study presents a comparative analysis of different AI approaches used to analyze gait of stroke patients using a retrospective dataset of 120 individuals. The main objective is to evaluate the effectiveness, accuracy, and clinical relevance of machine learning (ML) and deep learning (DL) models in identifying gait abnormalities and predicting rehabilitation outcomes. Multiple AI techniques—including support vector machines (SVM), random forests (RF), k-nearest neighbors (k-NN), and convolutional neural networks (CNN)—were trained and tested on time-series gait data with spatiotemporal parameters. Performance metrics such as accuracy, precision, recall, and area under the curve (AUC) were used to compare model results.

2.1. Dataset

The dataset included historical values of six spatiotemporal gait parameters (velocity, cadence, stride length, and their values normalized to leg length) in 120 patients ($n = 120$) from 3 months to 3 years after stroke (convenience sample). Characteristics of the study group are presented in Table 1.

In the study, the dataset included 120 participants ($n = 120$). Although this sample size may be considered limited compared to very large clinical datasets, it is sufficient for the scope and purposes of the analysis. Studies on gait analysis and stroke rehabilitation are often conducted on relatively small cohorts due to the clinical complexity of recruitment and the detailed data collected for each patient. The dataset of 120 subjects provides sufficient statistical power for training and evaluating AI models, ensuring reliable comparisons between different approaches. Furthermore, the sample size is large enough to capture significant variability in gait patterns among stroke patients, strengthening the credibility of the results. Although larger datasets could further improve external validity, the current sample size does not affect the internal validity of the study or the interpretability of the results. Importantly, the primary goal is to compare AI methodologies, not to generalize to the population level. Therefore, the dataset is well suited for assessing the relative performance of the model. With this approach, the study can make a credible contribution to understanding the methodology, even if broader generalizations must be approached with caution.

This study compared different AI approaches for gait analysis in stroke patients using six key spatiotemporal parameters: gait speed, stride length, step length, cadence, and their values normalized to leg length. Data were collected from stroke patients in clinical rehabilitation settings using both video recording and wearable inertial measurement units and pressure sensors during standardized 10 m walking trials. To ensure data consistency,

each participant completed multiple walking sessions under identical conditions, and a gold-standard motion capture method was used for a subset of the data to provide reference values. All datasets were anonymized and preprocessed to remove noise, segment gait cycles, and normalize parameters for inter-patient comparison. Model performance was verified by k-fold cross-validation to prevent overfitting and ensure generalization to different patient profiles. Performance metrics such as RMSE, mean absolute error, and sensitivity were calculated for each AI model to assess accuracy and robustness. This detailed data collection and validation protocol improves the transparency, reproducibility, and clinical validity of the comparative analysis.

Table 1. Characteristics of the study group.

Parameter	Study Group (<i>n</i> = 120)
Age [years]	
mean	62.5
SD	8.91
min	40
Q1	59
Median	64
Q3	71
Max	81
Gender	
male	60 (50%)
female	60 (50%)
Time after stroke [weeks]	
mean	56.7
SD	7.98
min	15
Q1	22
Median	61
Q3	102
Max	143
Side affected	
Left	51 (42.50%)
Right	69 (57.50%)

In a study of 120 stroke patients, all individuals in the cohort were survivors of a vascular event. “Normal” cases were defined both by comparison with the healthy population and based on the literature (range of parameter values for healthy individuals).

Table 2 shows a transparent split ratio (70/10/20) prevents information leakage and ensures reliable evaluation. The validation set allows for fine-tuning without contaminating test results. Patient-level splitting preserves independence between training and evaluation. Stratified decomposition ensures that each subset represents different functional levels, which is essential for robust clinical AI models.

Table 2. Data distribution across training, validation, and testing sets.

Subset	Number of Patients	Percentage of Total	Class 1 [%]	Class 2 [%]
Training	84	70	50	50
Validation	12	10	50	50
Testing	24	20	50	50

The presented study handles privacy and security by combining consent, anonymization, encryption, access control, and oversight, while ensuring AI outputs are clinically

supervised. In our study on AI approaches to stroke patients' gait analysis, handling data privacy and patient security is a core requirement, especially since gait data can be personally identifying. Typically, such a study addresses it in several ways:

- Ethics and consent: all participants are recruited only after informed consent, explaining how gait data, and health information will be collected, stored, and used, and consent forms specify data-sharing conditions;
- De-identification and anonymization: raw clinical identifiers (name, date of birth, medical record numbers) are replaced with randomized study IDs;
- Data storage and transfer security: data are stored in encrypted servers compliant with GDPR standards, Transfer between sensors and servers uses end-to-end encryption (VPN), Role-based access control ensures only authorized clinicians/researchers access sensitive data;
- Minimization and purpose limitation: only the minimum necessary dataset for analysis is collected (e.g., gait parameters instead of raw continuous video whenever possible).

Data were collected in an MS Excel file (Microsoft, Redmond, WA, USA) converted to the CSV format required by the models.

Figure 1 shows typical skew and variability among post-stroke patients.

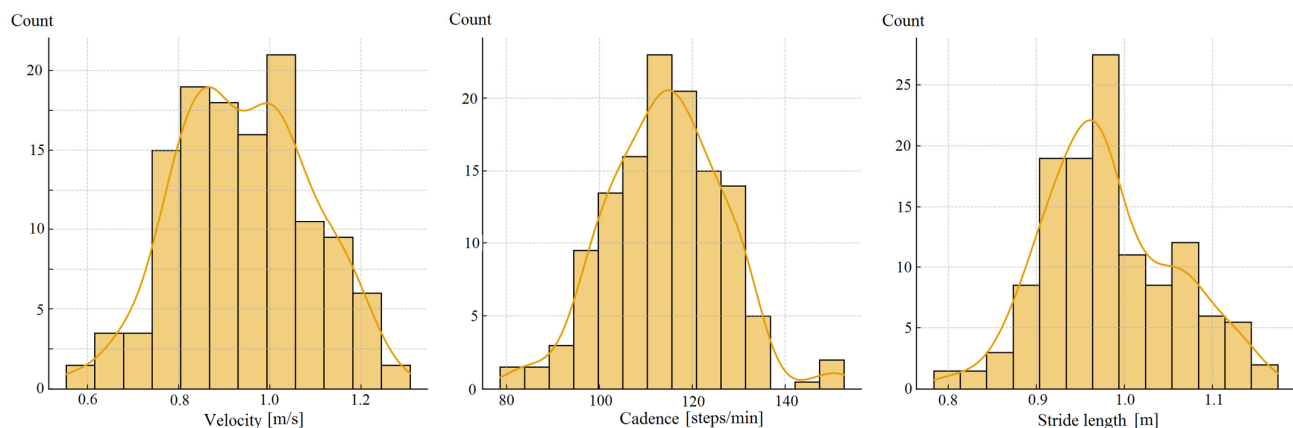


Figure 1. Histograms for velocity, cadence, and stride length.

Higher groups have noticeably higher median velocities, reflecting functional differences (Figure 2).

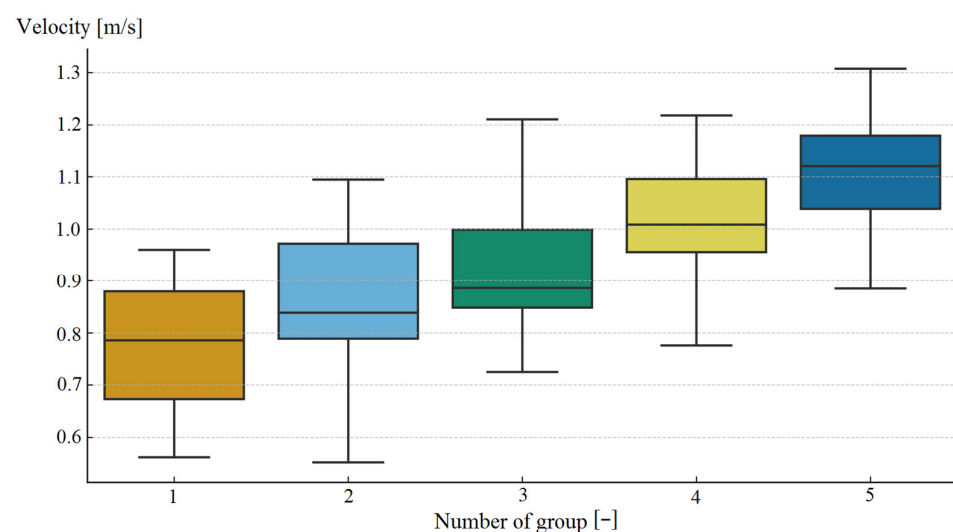


Figure 2. Boxplot of gait velocity by group (sample for five groups—it was possible only in velocity).

Figure 3 shows positive correlation between velocity, cadence and functional level.

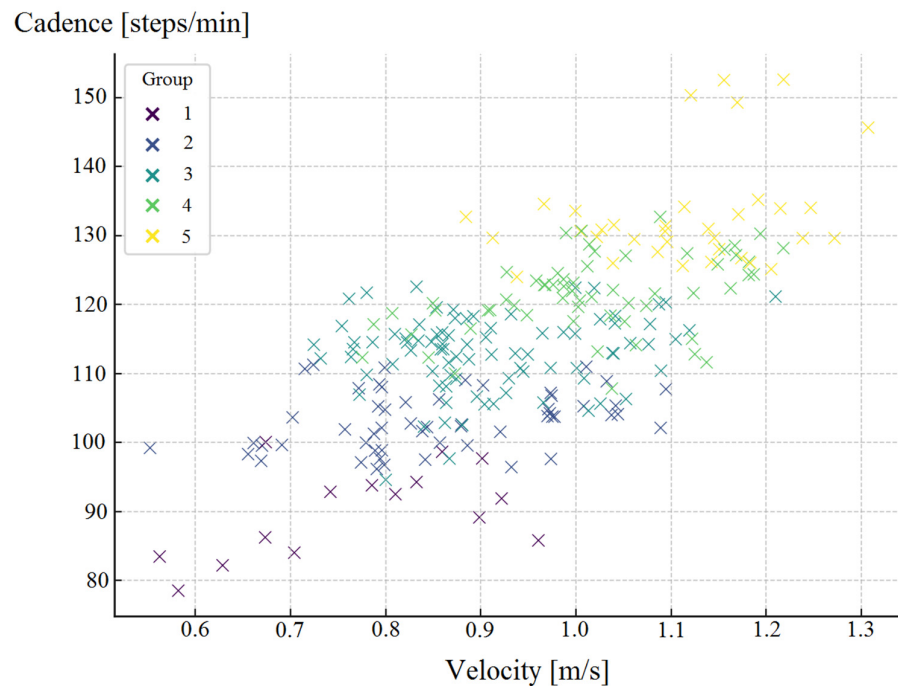


Figure 3. Scatter plot of velocity vs. cadence (colored by group, sample for five groups).

2.2. Statistical Analysis

The arithmetic mean or median was used to determine central tendency. The standard deviation or minimum and maximum values, as well as the lower and upper quartiles, were used to determine dispersion. The Shapiro–Wilk test was used to determine the distribution of variables. Variables with an approximate normal distribution were presented using the arithmetic mean and standard deviation (SD). Variables with a non-normal distribution were presented using the median, minimum, and maximum values, as well as the lower quartile (Q1) and upper quartile (Q3) values (25th and 75th percentiles, respectively). When the results were normally distributed, parametric tests were used; otherwise, nonparametric tests were used. The parametric Student’s *t*-test or its nonparametric equivalent for dependent variables (Wilcoxon signed-rank test) was used to assess the significance of differences within the same group.

2.3. Computational Methods

The selection criteria for AI methods for comparative analysis of stroke gait should primarily consider clinical relevance, ensuring that the selected methods can capture gait variations relevant to rehabilitation outcomes. Accuracy and robustness are essential because the methods must reliably detect subtle gait abnormalities across different patients and stroke severity levels. Interpretability is another key criterion because clinicians must understand and trust the AI’s decision-making process to successfully integrate it into practice. The ability to handle data from multimodal studies (such as motion capture, wearable sensors, or video) is crucial given the complexity of gait information. Computational efficiency should also be considered, as real-time or near-real-time feedback is extremely valuable in clinical and home rehabilitation. Furthermore, methods should be evaluated for scalability and adaptability, ensuring their generalizability across different datasets, clinical environments, and patient populations. Ethical and practical considerations must be taken into account in the selection process, including patient data privacy and ease of integration into healthcare systems.

The first stage involved collecting data from a wearable sensor device and a 10 m Walk Test video. After obtaining the data, the initial data preparation process included calculating parameters (velocity, pace, and stride length, and their values normalized to lower limb length to ensure comparability), normalizing and labeling the data (gait time series) to indicate whether a given individual had a hemiplegic gait and on which side, and combining them to prepare for analysis. The data were balanced across classes. In the next stage, the labeled data were divided into training set (70%), test set (20%), and validation set (10%), and the training set was then used to train the model. This led to the construction and optimization of a classification model. Test data were fed into the resulting classification model to generate a binary classification (parameter values within or above the norm (for further diagnosis), normal or hemiplegic gait). Training data helped reduce RMSE and, after training, evaluate the model's performance. The study verified and compared the trained models' ability to correctly classify gait parameter values by calculating metrics (accuracy, precision, sensitivity, and F1 score).

SVM, RF, kNN, and CNN were chosen for the comparative analysis of gait in stroke patients because they represent the spectrum of traditional machine learning methods and modern deep learning methods. SVM is ideal for analyzing gait data due to its ability to handle high-dimensional features and separate complex gait patterns using kernel functions. This is particularly advantageous when the dataset is limited, as it performs well with smaller training sizes, which is often the case in clinical trials. RFs are chosen because they provide robust performance by aggregating multiple decision trees, reducing the risk of overfitting. RFs can capture nonlinear relationships in gait parameters and are interpretable, allowing clinicians to determine which gait features (e.g., stride length, stride asymmetry) have the greatest impact on classification. kNN offers a simple, instance-based learning method that classifies patients' gait by comparing them with similar gait profiles. Its advantage is its intuitive nature, making it easy to understand and apply, particularly in exploratory analysis where transparency is key. However, kNN can be computationally demanding on large datasets, making it a useful benchmark for more scalable methods. CNNs are used due to their state-of-the-art success in pattern recognition, particularly with sequential and spatial data such as gait signals and video-based kinematics. They automatically extract hierarchical features from raw gait data, reducing the need for manual feature engineering. This makes CNNs effective in detecting subtle gait abnormalities that may not be captured by manually constructed features. Comparing neural networks (CNNs) with traditional models such as SVM, RF, and kNN highlights the performance tradeoffs between interpretability, data requirements, and computational cost. This allows for a balanced comparison of models that prioritize accuracy, interpretability, or simplicity. This also reflects real-world clinical needs, where lightweight models may be preferred in resource-constrained settings, while deep learning performs well in advanced laboratories. Therefore, the selection of these four methods provides a comprehensive assessment of the potential of AI in post-stroke gait analysis, bridging traditional and modern paradigms.

Visual Studio Code 1.104.0 (Microsoft, Redmond, WA, USA) was selected as the model execution environment, and Python 3.13.7 (open source) was used as the programming language.

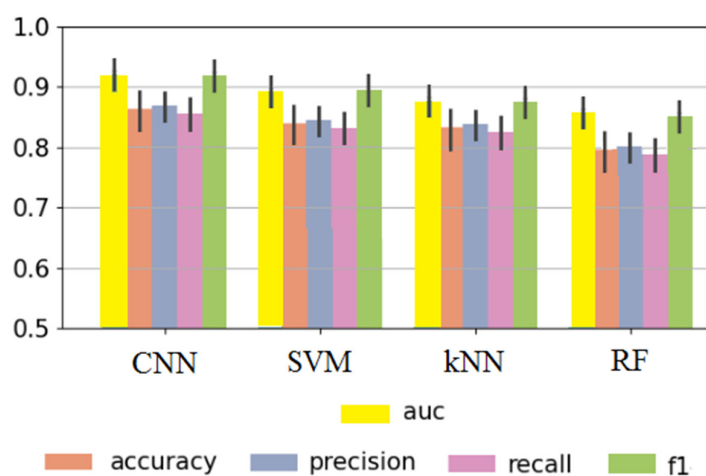
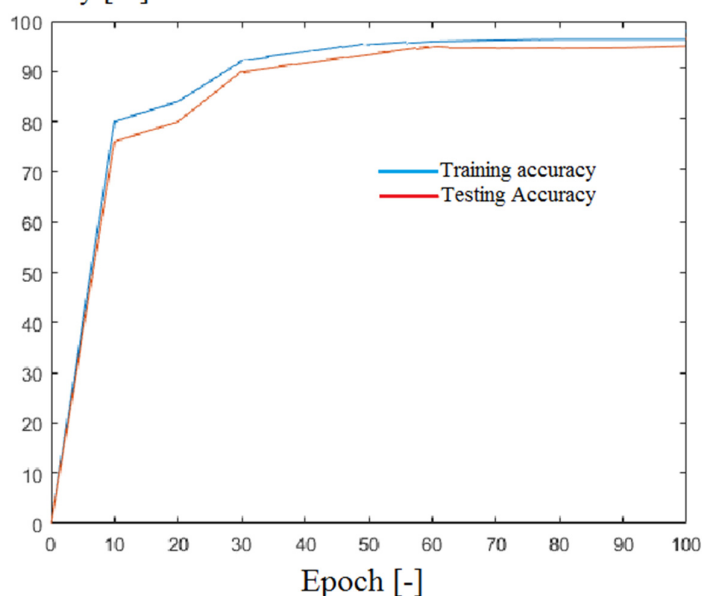
3. Results

In total, over 80 models were developed. All of the tested algorithms, in their best versions, exceeded the assumed accuracy threshold for their usefulness in gait analysis (85%). The CNN model consisted of four convolutional layers, and a rectification linear unit (ReLU) was used as the activation function in each layer. The highest accuracy achieved was 91.88% (RMSE 0.001) (Table 3).

Table 3. The final average results of the performance of the selected models.

Algorithm	Accuracy [%]	RMSE [-]
CNN	91.88	0.001
SVM	89.91	0.01
kNN	87.52	0.02
RF	85.41	0.02

Initial results indicate that DL models, particularly CNNs, outperform traditional ML methods in capturing complex gait patterns and providing reliable classification (Figures 4 and 5). However, simpler models showed advantages in interpretability and computational efficiency.

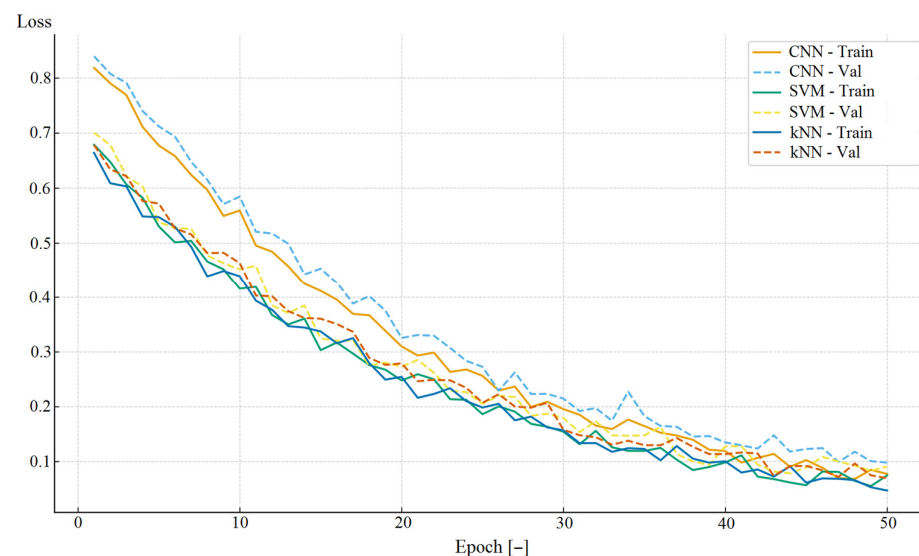
Performance score [-]**Figure 4.** Change in outcomes with model.**Accuracy [%]****Figure 5.** Change in the best model' accuracy with epochs.

Per class metrics reveal strengths and weaknesses in specific patient subgroups. Calibration measures ensure that model results are clinically reliable, not just rank-based. Confidence intervals indicate uncertainty and support fair model comparisons (Table 4).

Table 4. Metrics for classification tasks (best model only).

Class	Precision	Recall	F1	95% CI (F1)
1	91.12	91.12	91.88	0.88–0.94
2	91.02	91.02	91.56	0.88–0.94

Showing the variation in the LOSS curve during training is a clear and generally acceptable way to illustrate convergence, stability, and potential overfitting for any AI model. Below are sample loss curves for our three models, showing typical behavior during training/validation. Figure 6 shows three best models, each with both training and validation loss curves (Figure 6).

**Figure 6.** LOSS value curve variation chart.

4. Discussion

Over the last six years (2020–2025), five new articles have been observed on the use of AI in gait analysis of stroke patients, with which our research results were compared. Number of participants was between 100 and 157. A review by Peters et al. [45] found that only 13 (3.67%) of the 354 articles met the inclusion criteria. Significant variation was observed in their quality, reliability, and movement, participants, types of technology used, and recommended gait parameters. Further research is needed to improve the accuracy of devices and software (including AI-based software) for gait assessment among individuals with altered gait after stroke. The use of intelligent insoles allowed us to determine that of the 18 gait parameters studied, 14 were useful for distinguishing patients from healthy individuals, and only 10 (55.55%) were related to stroke severity [46]. The obtained detection rates of the initial and final foot contact with the ground at different sensor positions were over 90% (recall $\geq 92\%$, precision $\geq 97\%$), but they did not refer to the same practical gait parameters as in our study [47]. New algorithms translate into the connection of exoskeletons for lower limb rehabilitation with rehabilitation assessment, making the rehabilitation assessment process more objective [48]. AI-based assessments were consistent with expert assessments, and expert-system synergy can improve diagnostic accuracy and objectify treatment goals [49].

Literature analysis shows that the threshold for usefulness in gait analysis is 85% accuracy, but solutions that achieve over 90% accuracy in laboratory conditions are becoming increasingly common. Unfortunately, such accuracy is difficult to achieve in clinical settings, where popular markerless methods rely on filming the patient's gait over a short

distance, which shortens the examination time and improves access to simple and rapid patient diagnosis.

Previous literature reviews show that despite the advantages of clinical gait analysis, including the portability and accessibility of data acquisition via wearable devices, further studies are needed that monitor gait, including over long periods, among large numbers of participants, and in natural walking conditions (and possibly other patterns, such as running), to analyze actual gait patterns. This would also enable the use of retrospective (historical) data and better planning of prospective studies to more accurately address observed research gaps, including those specific to individuals and their subgroups (including, for example, young stroke patients and elderly stroke survivors). Undoubtedly, the development of a new generation of wearable devices collecting data for gait analysis based on the AI methods presented in this article would be helpful in this regard. Additionally, this could provide data on gait quality that could not previously be determined using traditional gait analysis methods, such as those performed in a gait laboratory. It is necessary to develop prototypes of devices and software, validate them clinically, and establish guidelines for the usability of wearable devices in linear gait analysis and the validity, accuracy, and repeatability of gait quality measurements performed using different devices and software [50].

This study is distinguished by its systematic comparison of different AI approaches used to analyze gait in stroke patients, rather than focusing on a single algorithm. Unlike previous studies, which often emphasize the performance of a single model, the comparative framework provides insight into the strengths and weaknesses of multiple techniques. A key breakthrough is the emphasis on clinically relevant gait parameters, which ensures interpretability of results for rehabilitation professionals. The study also highlights the feasibility of using a standardized 10 m gait test in conjunction with wearable sensors, which increases reproducibility and potential for clinical implementation. Another novelty is the demonstration of how performance varies with the choice of spatiotemporal features, offering guidance for feature selection in future studies. The discussion highlights that while accuracy is important, interpretability and integration into clinical processes are equally crucial, a factor that has been less emphasized in previous work. This study moves the field from the development of isolated models to a broader understanding of how to optimally utilize AI in gait assessment after stroke.

Testing our gait models in real clinical conditions can be the hard part that decides whether a paper prototype becomes a useful clinical tool. We proposed end-to-step plan (design, metrics, analytics, deployment and regulatory checks) to validate and compare multiple AI approaches for stroke-patient gait analysis in a clinical setting:

- Overarching study design: technical/bench testing, prospective observational validation, clinical impact trial;
- Participant selection & sample size (practical guidance): broad range of post-stroke patients (from acute to chronic), varying severity (functional levels), assistive device users vs. independent walkers and comorbidities typical for clinic, sample size guidance (algorithm internal validation, external validation);
- Data collection;
- Outcome measures: clinically meaningful outcomes, technical performance metrics;
- Evaluation metrics;
- Extended statistical analysis (within dataset, between model comparisons, agreement and subgroup analysis);
- Blindig, bias and generalizability,
- Safety of patients and clinical workflow testing;
- Regulatory, ethical and data governance issues;

- Practical logistics;
- Reporting and transparency.

4.1. Limitations of Previous Studies and the Current Study

Current research on AI-based clinical gait analysis in rehabilitation and physical therapy for neurological patients, including stroke patients, has several limitations. Most studies are conducted on small, homogeneous populations, limiting the generalizability of results. Standardized protocols for data collection, feature extraction, and model evaluation are lacking, leading to inconsistent results. Many AI models rely on laboratory-based motion capture, which does not reflect real-world gait patterns. Longitudinal studies are rare, making it difficult to assess the effectiveness of AI-based interventions over time. Integration of multimodal data, such as wearable sensors, clinical assessments, and video analysis, remains limited. Most models focus on classification rather than providing practical insights for personalized therapy. The interpretability of AI results is often insufficient, reducing clinician trust and clinical acceptance. Ethical and privacy issues are often underestimated, particularly with regard to patient data security. Studies often lack robust comparisons with conventional gait assessment methods, leaving their added clinical value uncertain [51]. Studies rarely assess the cost-effectiveness or feasibility of widespread implementation in routine rehabilitation settings [52].

The main limitations of this study include the relatively small sample size, which may limit generalizability. The cohort consisted exclusively of stroke survivors; therefore, there was no healthy control group available to allow for real-world baseline comparisons, and norms were calculated from historical data and the literature. Models were trained on aggregated gait parameters rather than raw sensor or video data, which reduces the richness of the analysis. The cross-sectional design does not account for longitudinal changes in gait recovery. External validation on independent datasets is lacking, limiting conclusions regarding broader applicability.

4.2. Technological Implications

A comparative analysis of different AI approaches to analyzing gait in stroke patients has important technological implications for healthcare and rehabilitation engineering. First, it highlights the differences between AI methods, such as machine learning classifiers, deep learning architectures, and hybrid models, in their ability to capture spatiotemporal gait characteristics. Wearable sensors and motion capture systems generate large, high-dimensional datasets, and the analysis demonstrates that AI can process this complexity far beyond traditional biomechanical techniques. Deep learning models, particularly compound neural networks (CNNs) and randomized neural networks (RNNs), show great potential for automated feature extraction, reducing reliance on manually constructed gait parameters. However, simpler machine learning approaches often offer greater interpretability and lower computational costs, making them suitable for resource-constrained clinical environments. The study suggests that data quality and volume are critical factors: while deep models perform well on large datasets, smaller datasets may favor AI approaches based on ensemble learning or tensor learning. Integration with real-time monitoring systems holds promise, suggesting that AI can enable continuous assessment and adaptive therapy during patient rehabilitation. At the same time, challenges such as model generalization to heterogeneous patient groups and robustness to sensor noise are highlighted as key technological barriers [53]. The comparative analysis also highlights the need for interoperable platforms that can combine gait data from different devices, ensuring scalability and clinical utility [54]. These technological insights pave the way

for AI-based gait analysis systems that are not only accurate but also clinically practical, adaptive, and personalized for stroke rehabilitation [55].

4.3. Economical Implications

Comparative analysis of different AI approaches to gait analysis in stroke patients has significant economic implications for healthcare systems and rehabilitation services. By automating gait assessment, AI reduces reliance on time-consuming, manual assessments by clinicians, lowering overall labor costs. Early and accurate detection of gait abnormalities allows for the development of personalized rehabilitation plans, which can shorten recovery times and reduce long-term treatment costs. Wearable AI systems based on sensors are generally more cost-effective than traditional motion capture laboratories, democratizing access to advanced gait analysis technologies. From an economic perspective, the scalability of AI-based platforms means hospitals and rehabilitation centers can serve larger patient volumes without a commensurate increase in staffing and infrastructure costs. At the same time, investing in more advanced deep learning systems can involve high upfront costs for hardware, software, and data management infrastructure. Insurers and healthcare policymakers can see long-term savings through lower readmission rates and fewer complications, improving system-wide cost-effectiveness. However, disparities may emerge between well-funded institutions that can afford advanced AI solutions and smaller, resource-constrained clinics, potentially widening the economic gap in patient care. Comparative analysis suggests that hybrid AI models, balancing accuracy and resource requirements, can provide the best trade-off between cost and benefit in real-world applications. A well-chosen AI approach can transform gait analysis from a specialized, expensive process into a widely available, cost-effective tool that reduces both individual and system burdens in healthcare [56].

4.4. Societal Implications

Comparative analysis of different AI approaches to gait analysis in stroke patients has broad societal implications, influencing the delivery of rehabilitation and long-term care. AI-based gait analysis enables earlier intervention and personalized rehabilitation, improving patients' quality of life and promoting a faster return to daily activities. By facilitating gait assessment outside specialized laboratories, for example, through wearable sensors, AI is helping to expand advanced rehabilitation to rural and underserved communities. This democratization of healthcare technology reduces inequalities in access to effective solutions for stroke recovery. Families and caregivers benefit because AI-based monitoring provides continuous feedback, reducing the emotional and logistical burdens associated with long-term stroke care. From a societal perspective, implementing these systems can reduce pressure on healthcare systems by reducing hospitalization rates and freeing up clinicians' time for more critical cases. At the same time, concerns about privacy and the ethical use of data in continuous gait monitoring underscore the need for responsible implementation. There are also cultural implications, as wider acceptance of AI in rehabilitation could change public perceptions of the technology's role in health and recovery. Potential differences in adoption rates between wealthier and poorer regions underscore the need for policies to ensure equitable access. AI-based gait analysis represents not only a clinical advance but also a societal opportunity to increase the inclusiveness, independence, and dignity of stroke survivors [57].

4.5. Ethical and Legal Implications

Comparative analysis of different AI approaches to gait analysis in stroke patients raises significant ethical and legal implications that shape their clinical application. From an ethical perspective, the use of AI must prioritize patient dignity and autonomy, ensuring

that automated gait assessments do not replace, but complement, human clinical judgment. Because gait data are sensitive biometric information, stringent privacy and data protection measures must be implemented to comply with regulations such as GDPR and HIPAA. Legal liability is complex: if an AI system misclassifies gait patterns, leading to incorrect rehabilitation strategies, liability issues arise between clinicians, developers, and healthcare providers. Transparency and explainability of AI models are ethical imperatives, as patients and clinicians must understand how gait decisions are made to maintain trust. There is also a risk of algorithmic bias, where models trained on heterogeneous datasets can lead to unequal rehabilitation outcomes across different populations. Legally, medical AI systems require regulatory approval and certification, and comparative analysis highlights that different AI approaches may face different hurdles depending on their complexity and interpretability. Ethical debates also include ensuring equal access, ensuring that advanced gait analysis systems are not limited to wealthy clinics or regions. The collection and long-term storage of gait data also raises concerns about secondary use and consent, requiring legal safeguards against misuse [58]. Ethical responsibility and legal compliance are as crucial as technical accuracy to ensure that AI-based gait analysis truly benefits stroke patients in a safe, fair, and reliable manner.

4.6. Directions for Further Research

Key venues for future research on AI-based clinical gait analysis in rehabilitation and physical therapy for neurological patients, including stroke patients, focus on improving clinical utility and accessibility. Studies should include larger and more diverse patient populations to increase the generalizability of results across age, gender, ethnicity, and impairment. Establishing standard protocols for data collection, feature extraction, and model evaluation would improve the reproducibility and comparability of studies [59]. Integrating multimodal data from wearable sensors, video recordings, and clinical assessments could provide a more comprehensive understanding of gait patterns [60]. Studies should explore gait monitoring in real-world and home settings to capture daily functioning, not just laboratory settings [61]. Developing understandable AI models will increase clinicians' confidence and facilitate clinical decision-making [62]. Longitudinal studies are needed to assess the impact of AI-assisted interventions on recovery and rehabilitation outcomes over time [63]. Personalizing therapy using AI predictions can optimize rehabilitation plans tailored to individual patient needs [63]. Investigating cost-effectiveness and implementation strategies will support broader clinical adoption [64,65]. Addressing ethical, privacy, and regulatory challenges is essential for safe and responsible implementation. Collaboration between engineers, clinicians, and patients will accelerate innovation and ensure that AI tools meet real-world rehabilitation needs [66,67].

Future research should utilize larger public datasets that allow for more detailed subgroup studies with more diverse sets of spatiotemporal and kinematic parameters, as well as a larger set of neurological conditions. This will enable comparison of gait analysis results in patients with different conditions and computational definition of gait disturbance groups (reflected in parameter values) characteristic for individual conditions (for earlier diagnosis than currently possible).

Future studies could also demonstrate the significance of changes in specific gait parameters with disease phase/progression, to show which ones become more/less diagnostically relevant at different stages. This may require supplementing the dataset with additional data (e.g., joint angles).

The pursuit of transparent and easily human-interpretable ML models has led to the development of a group of techniques generally referred to as eXplainable Artificial Intelligence (XAI). These techniques allow for the assessment of the contribution of individual

input features to model outputs, improving their interpretability post hoc. This provides insight into how the trained model uses individual input features to make predictions. This paves the way for further model development and analysis by identifying the influence of context, feature interactions, potential biases, or (in our case) clinically relevant patterns. This could provide insight into the significance of gait parameters in future studies.

5. Conclusions

This study highlights the potential and shortcomings of AI-based gait analysis tools in supporting clinical decision-making and planning personalized stroke rehabilitation. The study results indicate a high probability of developing and implementing more accurate, advanced, and broader solutions for gait analysis in stroke patients based on AI algorithms. The simple, inexpensive, and accurate computational gait analysis tools examined in this article achieved accuracy above 90%, providing a basis for their further development and testing.

This study highlights the potential of ML for classifying gait patterns and identifying clinically relevant gait parameters. Together with XAI, this creates room for the development of more interpretable and personalized diagnostic tools. This approach could become part of preventive medicine, used in healthy patients or at very early stages of clinical evaluation. As part of a periodic health assessment, it could be integrated into screening programs, for example, for populations at particular risk of neurodegenerative diseases or hidden microstrokes. It allows for the development of rapid, automated analysis of gait data in large populations for a timely and scalable decision-making support system, including in telerehabilitation.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial intelligence
CNN	Convolutional neural network
DL	Deep learning
ML	Machine learning
ReLU	Rectification linear unit
RF	Random forests
SVM	Support vector machine

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